INDUCED LORENTZIAN POLYNOMIALS: NOTES

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HOMEWORK AND ACTION ITEMS

- Is induction a volume polynomial preserver? [see here for volume preservers]✓ **DQ says:** And there's a nice presentation of $G^\vee$ and other operations.
- Write down a conjectured symbol or target polynomial whose multiaffine part generalizes the Dilworth $k = 1$ truncation.
- (DQ) Simplify and complete the symbol proof✓.

RUNNING COMMENTS

- **DQ says:** I don't know where to put this, but perhaps it is worth noting that the normalization on the codomain is necessary for Lorentzian preserving. The operators where you remove normalization on the target fails to be a Lorentzian preserver, i.e., $x^\alpha \mapsto \sum_{\beta \in \mathcal{J}_\alpha(\mathcal{S})} y^\beta$ or $N(x^\alpha) \mapsto \sum_{\beta \in \mathcal{J}_\alpha(\mathcal{S})} y^\beta$. An easy counter example is $\mathcal{S} = K_{2,2}$ and the image of $x_1 x_2$.
- **DQ says:** The set of representatives of $I \subseteq [m]$ is the support of a product of linear forms; for multisets, you get something similar but support of a product of elementary symmetrics.
- **DQ says:**

THINGS TO WORK ON

(1) **Dilworth truncations and k-dragons.** Another fun thought I had, which can be interesting to pursue, is the following "dragon" variation of our transversalization operator. Let me describe it in the case of matroids.

So far, in our case, we are counting whether a matching exists, i.e. using Hall's marriage condition, which reads: If $S_1, ..., S_n$ are subsets of $[n]$, then a matching exists if and only if $|\bigcup_{i \in I} S_i| \geq |I|$ for all subsets I of $[n]$.

Here is the "k-th dragon condition": A collection $S_1, ..., S_{n-k}$ of subsets of $[n]$ satisfies the "k-th dragon marriage condition" if it satisfies $|\bigcup_{i \in I} S_i| \geq |I| + k$ for all nonempty subsets of $[n-k]$. When $k = 1$, this is the dragon marriage condition that appears in Postnikov's "Permutohedra, Associahedra, and Beyond" paper (you can find why it's called "dragon" there). When k=3, this is called the Cerberus condition in my past paper with Brakensiek, Larson, Li.

Generally, these dragon conditions have close connection with Dilworth truncations. The first Dilworth truncation of the boolean matroid is the complete graph, whose basis generating polynomial is stable. I don't think there is much known about basis generating polynomials of higher Dilworth truncations.

Perhaps there is an operator with this version of marriage condition rather than our usual one, and it displays some interesting stability properties in certain situations?

(2) **Cogenerator rings.** Something one can also think about is the cogenerator ring of our transversalized polynomial. Without transversalizing, the old answer is the augmented Chow ring of the matroid. Can one find a matroid for which this new ring does not satisfy the whole Kähler package?

(3) **Mapping T_p to T_p.** Let's compose the induction operator $\mathcal{I}_{\mathcal{S}}$ with the "take multiaffine part" operator. Then, we can attempt to define a map $Gr_M(T_p) \to Gr_N(T_q)$ where N is the induced matroid from M along $\mathcal{S}$ and T_p and T_q are triangular hyperfields. Our theorem currently states that this map sends Lorentzian to Lorentzian. Does it send T_p-points to T_p-points? For which values of (p, q) is this map actually well-defined?

(4) **Recovering known polynomials.** What happens when you take a tame Lorentzian polynomial, say a monomial, and push it through $\mathcal{T}_{\mathcal{S}}$ multiple times (the same $\mathcal{S}$)? More generally, what known classes of Lorentzian polynomials can we recover by pushing through a good sequence of set systems?

(5) **Representability as transversalization.** Other "basic" Lorentzian preserving operators (of degree 0) are also "representable" by a choice of set system (e.g., something like sending a variable x to the linear form $x_1 + ... + x_m$, maybe with normalization). What other"basic" Lorentzian preserving operations are "representable"?

Main statement.

Let r be a positive integer, and let $\alpha \in \mathbb{Z}^n_{\geq 0}$ and $\beta \in \mathbb{Z}^m_{\geq 0}$ such that $|\alpha|_1 = \alpha_1 + \cdots + \alpha_n = r$ and $|\beta|_1 = \beta_1 + \cdots + \beta_m = r$. For a sequence $\mathcal{S} = (S_1, \ldots, S_m)$ of subsets of $[n] = \{1, \ldots, n\}$, we say that α *admits an $\mathcal{S}$-transversal to* β if there is a sequence of subsets $(T_1 \ldots, T_m)$ such that $T_i \subseteq S_i$ with $|T_i| = \beta_i$ for all $i = 1, \ldots, m$ and $\sum_{i=1}^m \mathbf{e}_{T_i} = \alpha$. Here $\mathbf{e}_T$ denotes the sum of standard basis vectors $\sum_{j \in T} \mathbf{e}_j \in \mathbb{Z}^m$. We denote by $R_\alpha(\mathcal{S})$ the set of all β such that α admits an $\mathcal{S}$-transversal to β.

The main statement is the following: *For a sequence $\mathcal{S} = (S_1, \ldots, S_m)$ of subsets of $[n]$ and a Lorentzian polynomial $\sum_{\alpha \in \mathbb{Z}^n} c_\alpha \frac{x^\alpha}{\alpha!}$ (of total degree r), the polynomial*

$$\sum_\alpha c_\alpha \Big(\sum_{\beta \in R_\alpha(\mathcal{S})} \frac{x^\beta}{\beta!} \Big)$$

is also Lorentzian.

For example, let M be a matroid of rank r on the ground set $[n]$, and consider its basis generating function $\sum_{B \text{ a basis of } \mathrm{M}} x^B$. Here x^B denotes $\prod_{j \in B} x_i$. Note that when $\alpha = \mathbf{e}_A$ for a subset $A \subseteq [n]$, saying that $\beta \in R_\alpha(\mathcal{S})$ is equivalent to saying that there is a partition $A_1 \sqcup \cdots \sqcup A_m$ of A such that $A_i \subseteq S_i$ and $|A_i| = \beta_i$ for all i. Thus, we have that

$$\sum_{B \text{ a basis}} \sum_{\beta \in R_{\mathbf{e}_B}(\mathcal{S})} \frac{x^\beta}{\beta!} = \sum_{\substack{\beta \in \mathbb{Z}^m_{\geq 0} \\ |\beta|_1 = r}} C_\beta(\mathrm{M}, \mathcal{S}) \frac{x^\beta}{\beta!}$$

is a Lorentzian polynomial, where $C_\beta(\mathrm{M}, \mathcal{S})$ is the number of bases B of M such that it admits a partition $B_1 \sqcup \cdots \sqcup B_m$ with $|B_i| = \beta_i$ and $B_i \subseteq S_i$ for all $i = 1, \ldots, m$.

Since taking the multi-affine part preserves Lorentzian-ness, in the setting above, we find that $\sum_{A \subseteq [m]} C_{\mathbf{e}_A}(\mathrm{M}, \mathcal{S}) x^A$ is also Lorentzian. This may have some interesting implications to mixing problems.

1. Proof via polystellahedral variety and dually Lorentzian polynomials

Let $\kappa \in \mathbb{Z}^n_{\geq 0}$, and let $k = |\kappa|_1 = \kappa_1 + \cdots + \kappa_n$. In what follows, the upshot is that there is a Chow ring $A^\bullet(X_\kappa)$ satisfying the Kähler package, and it has elements $h_S \in \mathcal{K}(X_\kappa) \subset A^1(X_\kappa)$ that have the intersection product as given in Proposition 1.1. Let X_κ be the *polystellahedral variety* of type κ, which is a k-dimensional (smooth toric projective) variety introduced in [Eur–Larson Intersection theory of polymatroids], and defined as follows: For each $i \in [n]$, let $V_i \simeq \mathbb{C}^{\kappa_i}$, and for $S \subseteq [n]$, let $V := \bigoplus_{i \in [n]} V_i \to \bigoplus_{j \in S} V_j$ be the projection map. We define

$$X_\kappa := \text{ the closure of the image of } V \to \prod_{S \subseteq [n]} \mathbb{P}\Big(\mathbb{C} \oplus \bigoplus_{i \in S} V_i\Big).$$

Let $h_S \in A^1(X_\kappa)$ denote the hyperplane class pulled back from the factor $\mathbb{P}(\mathbb{C} \oplus \bigoplus_{i \in S} V_i)$. When S is a singleton $\{i\}$, we simply write h_i for $h_{\{i\}}$. The key fact about these divisors h_S on X_κ established in [Theorem 1.3 & Lemma 5.2, EL'24] is the following.

Proposition 1.1. Recall $k = |\kappa|_1$. For a sequence $\mathcal{S} = (S_1, \dots, S_k)$ of subsets of $[n]$, we have

$$\int_{X_\kappa} h_{S_1} \cdots h_{S_k} = \begin{cases} 1 & \text{if } \mathcal{S} \text{ has a matching to } \kappa \\ 0 & \text{otherwise,} \end{cases}$$

where "$\mathcal{S}$ has a matching to κ" means that there is a map $\iota : [k] \to [n]$ such that $\iota(i) \in S_i$ for all $i \in [k]$ and $\sum_{i \in [k]} \mathbf{e}_{\iota(i)} = \kappa$.

Note that the proposition specializes in the following way.

Corollary 1.2. Let α be a monomial of total degree r and multi-degree at most κ. Let $\mathcal{S} = (S_1, \dots, S_m)$ be a sequence of subsets of $[n]$, and let $\beta \in \mathbb{Z}^m_{\geq 0}$ with $|\beta|_1 = r$. Then, we have

$$\int_{X_\kappa} \Big(\prod_{i \in [m]} h_{S_i}^{\beta_i} \Big) \cdot (h_1^{\kappa_1 - \alpha_1} \cdots h_n^{\kappa_n - \alpha_n}) = \begin{cases} 1 & \text{if } \beta \in R_\alpha(\mathcal{S}) \\ 0 & \text{otherwise,} \end{cases}$$

where $R_\alpha(\mathcal{S})$ is the set of β such that α admits an $\mathcal{S}$-transversal to β (i.e. there is a sequence of subsets $(T_1, \dots, T_m)$ such that $T_i \subseteq S_i$ and $|T_i| = \beta_i$ for all $i \in [m]$ satisfying $\sum_{i \in [m]} \mathbf{e}_{T_i} = \alpha$).

We now recall that a polynomial $s^\vee(x_1, \dots, x_n)$ of multi-degree at most κ is *dually Lorentzian* if $s(x_1, \dots, x_n) := N(x^\kappa s^\vee(x^{-1}))$ is Lorentzian. The following is a reinterpretation of [Theorem 7.2, Ross–Süss–Wannerer].

Proposition 1.3. Let $s^\vee(x_1, \dots, x_n)$ be a dually Lorentzian polynomial of total degree $k - r$. If $(A^\bullet, \int, \mathcal{K})$ is a triple that satisfies the mixed Hodge–Riemann relations in degree at most 1 (i.e. $A^\bullet = \bigoplus_{i=1}^k A^i$ is a graded $\mathbb{R}$-algebra, $\int : A^k \xrightarrow{\sim} \mathbb{R}$ is the degree map, and $\mathcal{K} \subset A^1$ is the cone for which $\mathrm{HR}^{\leq 1}$ holds), then for any $D_1, \dots, D_m, h_1, \dots, h_n \in \overline{\mathcal{K}}$, the polynomial

$$\int \Big(\sum_{i=1}^n x_i D_i \Big)^r \cdot s^\vee(h_1, \dots, h_n)$$

is Lorentzian.

Now fix a Lorentzian polynomial $s(x) = \sum_\alpha c_\alpha \frac{x^\alpha}{\alpha!} \in \mathbb{R}[x_1, \dots, x_n]$ of total degree r and multi-degree at most κ. By definition, $s^\vee(x) = \sum_\alpha c_\alpha x^{\kappa - \alpha}$ is dually Lorentzian. Combining the corollary above with the proposition yields the following.

Theorem 1.4. For a sequence $\mathcal{S} = (S_1, \dots, S_m)$ of subsets of $[n]$, we have

$$\int_{X_\kappa} \Big(\sum_{i \in [m]} x_i h_{S_i} \Big)^r \cdot s^\vee(h_1, \dots, h_n) = \sum_\alpha c_\alpha \Big(\sum_{\beta \in R_\alpha(\mathcal{S})} \frac{x^\beta}{\beta!} \Big),$$

and in particular, the right-hand-side is a Lorentzian polynomial.

2. More generally, Transversalization is Lorentzian preserving

Observe that in the proofs above, the key idea was to refine the transversalized basis generating polynomial (TBGP) as a sum of some polynomials indexed by bases. This linear operation of *transversalizing* or *inducing* a Lorentzian polynomial along a set system $\mathcal{S}$ turns out to be Lorentzian preserver. This subsumes the Lorentzian property of TBGPs.

Let $\mathcal{S} = (S_1, \ldots, S_n)$ be a set system on $[m]$ and fix $d \in \mathbb{Z}_{>0}$. Then, for any $\alpha \in \mathbb{Z}^m_{\geq 0}$ and $\beta \in \mathbb{Z}^n_{\geq 0}$ with $|\alpha| = |\beta| = d$, we say that α *admits an $\mathcal{S}$-transversal to* β if there exists a system of subsets $\mathcal{T} = (T_1, \ldots, T_n)$ with $|T_i| = \beta_i$ and the multiset union of $\mathcal{T}$ is α. For fixed $\alpha \in \mathbb{Z}^m_{\geq 0}$, we call the set of elements $\beta \in \mathbb{Z}^n_{\geq 0}$ such that α admits an $\mathcal{S}$-transversal to β the *image of* α or *collection of representatives of* α *along* $\mathcal{S}$:

$$R_\alpha(\mathcal{S}) := \{\beta \in \mathbb{Z}^n_{\geq 0} : \alpha \text{ admits an } \mathcal{S}\text{-transversal to } \beta\}.$$

Let $g_\alpha(\mathcal{S}) \in \mathbb{R}[y_1, \ldots, y_n]$ be the exponential generating function of $R_\alpha(S)$, i.e.,

$$g_\alpha(\mathcal{S}) = \sum_{\beta \in R_\alpha(\mathcal{S})} \frac{y^\beta}{\beta!}.$$

Now we are equipped to define the *transversalization along* $\mathcal{S}$:

$$\begin{aligned} T_{\mathcal{S}} : \mathbb{R}[x_1, \ldots, x_m] &\longrightarrow \mathbb{R}[y_1, \ldots, y_n], \\ x^\alpha \quad &\longmapsto \quad g_\alpha. \end{aligned}$$

We will denote $T^\kappa_{\mathcal{S}}$ as the restriction of $T_{\mathcal{S}}$ onto the polynomial ring of bounded degree κ. As noted by Eur, this operator is a Lorentzian/polymatroidal analog to the "induced matroid along a given set system" as found in [Chapter 8.2; Welsh76].

In [Brändén-Huh], we are given a sufficient condition for linear and homogeneous operators that preserve Lorentzian polynomials.

Theorem 2.1. *Let T be a homogeneous operator of degree ℓ on $T : \mathbb{R}_\kappa[x_1, \ldots, x_m] \to \mathbb{R}_\gamma[y_1, \ldots, y_n]$ and $|\kappa| < \infty$. If $sym_T(y, u)$ is Lorentzian, then for all Lorentzian f, $T(f)$ is Lorentzian.*

We can immediately compute the symbol:

$$sym_{T_{\mathcal{S}}}(y, u) = \sum_{0 \leq \alpha \leq \kappa} \binom{\kappa}{\alpha} g_\alpha u^{\kappa - \alpha}.$$

It suffices to prove that is Lorentzian.

2.1. Transversalized matroid statistic generating functions.

We first restrict our attention to the case when the domain is multiaffine.

Observation 2.1 (Input is a matroid statistic generating function)**.** We can restrict the action to polynomials supported on matroids. In this case, $\kappa = \mathbf{1} = (1, \ldots, 1) = [m]$. Then, we have

$$sym_{T^{\mathbf{1}}_{\mathcal{S}}}(w, u) = \sum_{I \subseteq [m]} g_I u^{[m] \setminus I} = \sum_{I \subseteq [m]} \sum_{\beta \in R_I(\mathcal{S})} \frac{y^\beta}{\beta!} \cdot \frac{u^{[m] \setminus I}}{I!}.$$

In particular, note that this is the exponential generating function of a set $\mathcal{J}_1(\mathcal{S}) \subseteq \{0, 1\}^m \times \mathbb{Z}^n_{\geq 0}$. So it suffices to prove that $\mathcal{J}_1(\mathcal{S})$ is an M-convex set. Observe that the set $\mathcal{J}_1(\mathcal{S})$ admits two natural

projections; we let π_m denote projecting away the first m coordinates and π_n denote projecting away the last n coordinates. This can be summarized in a diagram:

$$\begin{tikzcd} & \mathcal{J}_1(\mathcal{S}) \arrow[dl, "\pi_n"'] \arrow[dr, "\pi_m"] & \\ \{I^c \subseteq [m]\} & & R(\mathcal{S}) \end{tikzcd}$$

where $R(\mathcal{S}) \subseteq \mathbb{Z}^n$ is the independent sets of the polymatroid induced by the rank function

$$rk_{\mathcal{S}} : 2^n \to \mathbb{Z}, \quad J \mapsto \left| \bigcup_{j \in J} S_j \right|.$$

Remark 2.2. As we aim to prove that $T^{\mathbf{1}}_{\mathcal{S}}$ is a Lorentzian preserver, i.e., Lorentzian morphism, one should also (at least loosely) think of $\mathcal{J}_1(\mathcal{S})$ as a morphism between $M-$convex sets. One pulls back data along π_n and pushes forward along π_m.

By Theorem 2.1, to obtain the desired result on matroid statistic generating functions, it suffices to prove the following proposition:

Proposition 2.3. For any set system $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$, $\mathcal{J}_1(\mathcal{S})$ is an M-convex set.

Proving this gives us the following immediate corollary:

Corollary 2.4. For any set system $\mathcal{S}$ on $[m]$, transversalization is a Lorentzian preserver.

In particular, we recover the Lorentzian property of the transversalization of the matroid base generating polynomial, homogenized independence polynomial, and their valuated analogs.

2.1.1. *Pushing matroids through bipartite graphs.* Before proving the key Theorem 2.3, we recall a useful perspective inspired after reading recent work of Brandenburg, Loho, and Smith [BLS25]; these ideas of induction along a bipartite graph were guided and inspired by Eur via email.

Recall that every set system $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$ is equivalent to a bipartite graph $G_{\mathcal{S}} = ([m], [n]; \mathcal{E}_{\mathcal{S}})$ where $[m]$ is the ground set, $[n]$ is the indices of the S_i's and for each element in $j \in S_i$, we include the edge $(j, i) \in [m] \times [n]$. Moreover, without loss of generality, we can take $[m] = \bigcup_{i \in [n]} S_i$ and $S_i \neq \varnothing$ for all i. As a bipartite graph, this means that $G_{\mathcal{S}}$ can be taken without isolated vertices. To avoid notational clutter, when clear from context, we abuse notation and identify $\mathcal{S}$ with $G_{\mathcal{S}}$.

Next, observe $I \subseteq [m]$ admits a transversal along $\mathcal{S}_\beta$ precisely where there exists of a partial matching $\mu \subseteq \mathcal{E}_{\mathcal{S}}$ whose left vertices are I and right vertices are β_i many S_i's. To clarify the picture, we will define a class of subgraphs, effectively degenerate matchings, that capture our desired transversals with multiplicity.

For $I \subseteq [m]$, an *I-motching of type β on $\mathcal{S}$*[1] is a subgraph $\mu \subseteq \mathcal{E}_{\mathcal{S}}$ on vertex set $(I, supp(\alpha)) \subseteq [m] \times [n]$ and choice of edges such that vertices have the following degree conditions:

$$\text{for } v \in I, \deg_\mu(v) = 1 \text{ and for } i \in [n], \deg_\mu(S_i) = \beta_i.$$

[1]The name *I-motching* is a portmanteau of 'botched' and 'matching.'

The following examples take place on $G_{\mathcal{S}}$ where $\mathcal{S} = (12, 123)$.

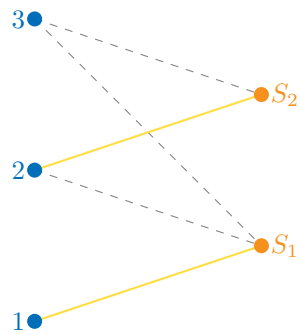

FIGURE 1. $\{1, 2\}$-matching of type $(1, 1)$

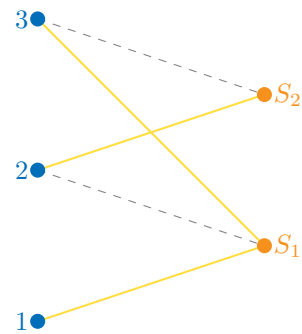

FIGURE 2. $[3]$-matching of type $(2, 1)$

Note that every I-matching of type β comes from a (partial) matching $I \to S_\beta$; conversely, every partial matching from I to S_β can be made into an $I-$motching of type β. This allows us to restate $\mathcal{J}_1(\mathcal{S})$ as the following set:

$$\mathcal{J}_1(\mathcal{S}) = \{(I^c, \beta) \in \{0, 1\}^m \times \mathbb{Z}^n : \text{there exists an } I\text{-matching of type } \beta\}.$$

Note that $\mathcal{J}_1(\mathcal{S})$ is similar to $\widetilde{\Gamma_{\mathcal{S}}}$ of [Example 3.8; BLS] with the crucial difference that $\mathcal{J}_1(\mathcal{S})$ consists of a subset of collections of edges whereas $\widetilde{\Gamma_G}$ runs over all possibles collections of edges. In any case, we claim that $J(\mathcal{S})$ is a something like a *linking set* in the sense of [BLS]. A classical result from discrete convex analysis is the characterization of M-convex sets as submodular functions. This can be found in Murota's text [§4.4 & §4.8; Mur03]. Some useful notation we follow: for $x \in J \subseteq \mathbb{Z}^E$ and $A \subseteq E$, let $x(A)$ denote the coordinate sum of x supported on A, i.e.,

$$x(A) = \sum_{i \in A} x_i.$$

Theorem 2.5. Let $J \subseteq \mathbb{Z}^E$. Then, J is M-convex if and only if the function $rk_J(A) = \max\{x(A) : x \in J\}$ is submodular.

By Theorem 2.5, we aim to prove that the rank function $rk_{\mathcal{J}_1(\mathcal{S})}$ is submodular. This will suffice since as a maximizer, it is immediately monotone and normalized.

2.1.2. *$rk_{\mathcal{J}_1(\mathcal{S})}$ is submodular.* The aim of this section is to obtain a concrete description of $rk_{\mathcal{J}_1(\mathcal{S})}$ from which submodularity is immediate. In particular, we aim to understand $\max\{x(J, T) : x \in \mathcal{J}_1(\mathcal{S})\}$ for all $(J, T) \subseteq [m] \times [n]$. In what follows, it will be useful to think of what is happening on restrictions of $G_{\mathcal{S}}$ onto (J, T). Precisely, let $G_{\mathcal{S}}(J, T)$ be the induced subgraph of $G_{\mathcal{S}}$ on the subset of vertices $(J, T) \subseteq [m] \times [n]$. Additionally, say $T = \{t_1 < \cdots < t_k\} \subseteq [n]$, let $\mathcal{S}(J, T)$ be the subset subsystem of $\mathcal{S}$ where

$$\mathcal{S}(J, T) := (S_{t_1}|_J, \cdots S_{t_k}|_J).$$

The following lemma will prove useful in the following analysis of $rk_{\mathcal{J}_1(\mathcal{S})}$.

Lemma 2.6. Let $(J, T) \subseteq [m] \times [n]$. Then

(a) $G_{\mathcal{S}}(J,T) = G_{\mathcal{S}(J,T)}$.
(b) For every I-motching of type β on $G_{\mathcal{S}}$, there exists a $(I \cap J)$-motching of type β_T on $G_{\mathcal{S}}(J,T)$ where $\beta_T \in \mathbb{Z}^T$ is the restriction of β by projecting away coordinates $[n] \setminus T$.

Proof. Proof of (a). Immediately the vertex sets agree. So it suffices to check that the edge sets agree. Take an edge $(j,i) \in G_{\mathcal{S}}(J,T)$, i.e., $j \in J$, $t \in T$. Since, $(j,t) \in G_{\mathcal{S}}$, this implies that $j \in S_t|_J$, i.e., $(j,i) \in G_{\mathcal{S}(J,T)}$. On the other hand, taking the induced subgraph guarantees that for every such $j \in S_t \cap J$, the corresponding edge (j,t) is in $G_{\mathcal{S}}(J,T)$, i.e., $G_{\mathcal{S}(J,T)} \subseteq G_{\mathcal{S}}(J,T)$.
Proof of (b). Let μ be an I-motching of type β on $G_{\mathcal{S}}$. Delete the edges appearing that share a vertex not in J. Removing edges preserves the (partial) motching property. The surviving edges are adjacent to $I \cap J$ in $[m]$ and has multiplicity given by β_T. □

To begin the analysis of $rk_{\mathcal{J}_1(\mathcal{S})}$, consider the following lemma/observation in the case when $A = [m] \times [n]$ is the full support of $\mathcal{S}$.
Lemma 2.7. Take some set system $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$. Then,

$$rk_{\mathcal{J}_1(\mathcal{S})}([m] \times [n]) = \max\{x([m],[n]) : x \in \mathcal{J}_1(\mathcal{S})\} = m.$$

In fact, $x([m],[n]) = m$ for all $x \in \mathcal{J}_1(\mathcal{S})$.

Proof. Any point $x = (I^c, \beta) \in \mathcal{J}_1(\mathcal{S})$ corresponds to $\mu \subseteq G_{\mathcal{S}}$ which is an I-motching of type β on $\mathcal{S}$. In particular, note that $|\beta| = |I|$ since μ is an I-motching. Then, $x([m],[n]) = |\beta| + |I^c| = m$. □

To prepare the full statement, we demonstrate two special cases which suggests the generalization to the correct concrete formula. First, consider when the second coordinate is empty:
Lemma 2.8. Let $J \subseteq [m]$. Then,

$$rk_{\mathcal{J}_1(\mathcal{S})}(J, \varnothing) = \max\{x(I, \varnothing) : x \in \mathcal{J}_1(\mathcal{S})\} = |J|.$$

Proof. In this case, the coordinate sum can only be nonzero on vertices from $J \subseteq [m]$. The weight on each such vertex is either 0 or 1 since $\mathcal{J}_1(\mathcal{S}) \subseteq \{0,1\}^m \times \mathbb{Z}^n$. Therefore, $x(J, \varnothing) \leq |J|$. Note that for any $\mathcal{S}$, the empty subgraph $\varepsilon \subseteq G_{\mathcal{S}}$ exists. Moreover, the empty graph ε is an $\varnothing$-motching of type $\mathbf{0}$. Since no vertices of J appear, we have that $\varepsilon(J, \varnothing) = |J|$. □

For the second case, consider when the first coordinate is empty:
Lemma 2.9. Let $T \subseteq [n]$. Then,

$$rk_{\mathcal{J}_1(\mathcal{S})}(\varnothing, T) = \max\{x(\varnothing, T) : x \in \mathcal{J}_1(\mathcal{S})\} = \left| \bigcup_{t \in T} S_t \right|.$$

Proof. In this case, the coordinate sum is restricted onto subsets S_t for $t \in T$; we let all other such sets vanish. All elements of $\mathcal{J}_1(\mathcal{S})$ correspond to a motching, and any such motching can at most use all the distinct vertices in $[m]$ that are adjacent to $t \in T$. That set of vertices is precisely $S_T := \bigcup_{t \in T} S_t$. Thus, the coordinate sum supported on $(\varnothing, T)$ is at most the size of that set, i.e.,

$$rk_{\mathcal{J}_1(\mathcal{S})}(\varnothing, T) \leq \left| \bigcup_{t \in T} S_t \right|.$$

To obtain equality, it suffices to construct a motching that uses all vertices in S_T. First, take the induced subgraph $G = G_{\mathcal{S}}([m], T)$; in particular, it is a subgraph of $G_{\mathcal{S}}$. If it is a motching, we are done. If it is not, there is some subset of vertices $V \subseteq [m]$ that where $\deg_G(v) \geq 2$ for $v \in V$. On each $v \in V$, trim off excess edges to reduce the degree to exactly 1. What remains will be an S_T-motching of some type γ where $|\gamma| = |S_T|$. In particular, $y := (S_T, \gamma) \in \mathcal{J}_1(\mathcal{S})$ and $y(\varnothing, T) = |S_T|$. □

Note that the rank function $rk_{\mathcal{J}_1(\mathcal{S})}$ must subsume both cases. This leads to the following guess:

Proposition 2.10. Let $(J, T) \subseteq [m] \times [n]$. Then,

$$rk_{\mathcal{J}_1(\mathcal{S})}(J, T) = \left| \bigcup_{t \in T} S_t \cup J \right|.$$

Proof. By Theorem 2.6, every I-motching of type β on $G_{\mathcal{S}}$ restricts to an $(I \cap J)$-motching of type β_T on $G_{\mathcal{S}}(J, T) = G_{\mathcal{S}(J,T)}$. Therefore, for any point $x = ([m] \setminus I, \beta) \in J(\mathcal{S})$, there exists a point $\overline{x} = (J \setminus (I \cap J), \beta_T) \in \mathcal{J}_1(\mathcal{S}(J, T))$. On the other hand, every I-motching of type β on $G_{\mathcal{S}(J,T)}$ embeds as an I-motching of type β on $G_{\mathcal{S}}$. Therefore, for any point $\overline{x} = (J \setminus I, \beta) \in \mathcal{J}_1(\mathcal{S}(S, T))$, there exists a point $x = ([m] \setminus I, \beta) \in \mathcal{J}_1(\mathcal{S})$. In both cases, observe that by by construction

$$x(J, T) = \overline{x}(J, T).$$

In particular, this implies that

$$rk_{\mathcal{J}_1(\mathcal{S})}(J, T) = rk_{\mathcal{J}_1(\mathcal{S}(J,T))}(J, T).$$

The right hand side is precisely in the setting of Theorem 2.7, i.e., of full support, so we obtain

$$rk_{\mathcal{J}_1(\mathcal{S})}(J, T) = \left| \bigcup_{t \in T} S_t \cup J \right|.$$

□

From this presentation, submodularity of the rank function is more or less immediate:

Proposition 2.11. The rank function $rk_{\mathcal{J}_1(\mathcal{S})}$ is submodular.

Proof. This is a consequence of inclusion exclusion on sets. Let (J, T) and (J', T') be subsets of $[m] \times [n]$. Then, we want to show that

$$\left| \bigcup_{t \in T} S_t \cup J \right| + \left| \bigcup_{t \in T'} S_t \cup J' \right| \geq \left| \bigcup_{t \in T \cup T'} S_t \cup J \cup J' \right| + \left| \bigcup_{t \in T \cap T'} S_t \cup (J \cap J') \right|.$$

Let $A = \bigcup_{t \in T} S_t$ and $A' = \bigcup_{t \in T'} S_t$. Note that $A \cup A' = \bigcup_{t \in T \cup T'} S_t$ and $A \cap A' = \bigcup_{t \in T \cap T'} S_t$. So it suffices to prove that

$$|A \cup J| + |A' \cup J'| = |(A \cup A') \cup (J \cup J')| + |(A \cap A') \cup (J \cap J')|.$$

A quick Venn-diagram check confirms this statement holds. □

So we obtain our desired results as immediate corollaries.

Proof of Theorem 2.3. This is a consequence of Theorem 2.5. □

Proof of Theorem 2.4. This is a consequence of Theorem 2.3 and [Theorem 3.2; BH]. □

2.2. **Transversalized generating functions on M-convex sets.** Using standard tricks appearing in the Borcea-Brändén theory on stable polynomials, we prove the following:

Theorem 2.12. For any choice of set system $\mathcal{S}$, transversalization $T_{\mathcal{S}}$ is a Lorentzian preserver.

The key idea and proposition to obtain the desired result is that, for fixed $\kappa \in \mathbb{Z}^m_{\geq 0}$, $T^\kappa_{\mathcal{S}}$ factors through polarization. Before stating the proposition, we recall the *polarization operator* which is defined as follows:

$$\begin{aligned} \Pi^{\uparrow}_{\kappa} : \mathbb{R}_\kappa[x_1, \ldots, x_m] &\longrightarrow \mathbb{R}[x_{1,1}, \ldots, x_{1,\kappa_1}, \ldots, x_{m,1}, \ldots, x_{m,\kappa_m}]_{MA} \\ x^\alpha &\longmapsto \binom{\kappa}{\alpha}^{-1} e_\alpha(x_\kappa) \end{aligned}$$

where $e_\alpha(x_\kappa) = \prod_{i\in[m]} e_{\alpha_i}(x_{i,1}, \ldots, x_{i,\kappa_i})$. We often use the label x_κ as a placeholder for the box of $x_{1,1}, \ldots, x_{m,\kappa_m}$ variables. Notice that the choice of normalization is such that the following holds:

$$\Pi^{\downarrow}_{\kappa} \circ \Pi^{\uparrow}_{\kappa} = id$$

where $\Pi^{\downarrow}_{\kappa}$ is the contraction map:

$$\begin{aligned} \Pi^{\downarrow}_{\kappa} : \mathbb{R}[x_\kappa]_{MA} &\longrightarrow \mathbb{R}_\kappa[x_1, \ldots, x_m] \\ x_{i,j} &\longmapsto x_i. \end{aligned}$$

Morally, one should think of polarization map as equidistributing the mass on non-squarefree terms on multiaffine lifts such that projection recovers the polynomial. The pair of maps enjoys the following crucial property [Proposition 3.1; BH]:

Proposition 2.13. For all κ, the operators $\Pi^{\uparrow}_{\kappa}$ and $\Pi^{\downarrow}_{\kappa}$ are Lorentzian preservers.[2]

Next, we introduce the set system $\mathcal{S}(\kappa)$, equivalently the graph $G_{\mathcal{S}(\kappa)}$, which we call the *polarization of* $\mathcal{S}$ along κ.[3]

Definition 2.14. Given a set system $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$, the *polarization of* $\mathcal{S}$ along κ is the set system on $[m_\kappa] := \{1_1, \ldots, 1_{\kappa_1}, \ldots, m_1, \ldots, m_{\kappa_m}\}$ consists of sets S^κ_j for $j \in [n]$ where whenever $i \in S_j$, for all $t \in [\kappa_i]$, $t \in S^\kappa_j$.

We can also state this equivalently as a graph operation. For any $\kappa \in \mathbb{Z}^m_{\geq 0}$, define $G_{\mathcal{S}(\kappa)}$ to be a bipartite graph on partite sets $[m_\kappa]$ and $[n]$ where for $i_t \in [m_\kappa]$, (i_t, j) is an edge in $G_{\mathcal{S}(\kappa)}$ if and only if $i \in S_j$. We demonstrate an example for clarity below:

[2]In fact, they are stability preservers. If we prove that $T^{\mathbf{1}}_{\mathcal{S}}$ is a stability preserver, we win.

[3]We do this for the sake of linguistic determination.

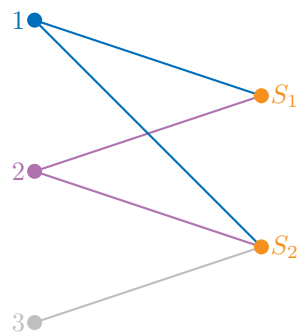

FIGURE 3. $G_{\mathcal{S}}$ where $\mathcal{S} = (12, 123)$

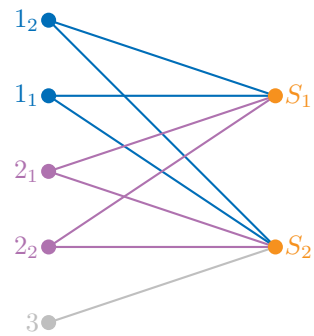

FIGURE 4. $G_{\mathcal{S}(2,2,1)}$ where $\kappa = (2, 2, 1)$

As hinted earlier, polarization and transversalization–a priori, surprisingly–play well together. This is the content of the following proposition:

Proposition 2.15. For any $\kappa \in \mathbb{Z}^m_{\geq 0}$, choice of set system $\mathcal{S}$,

$$T^{\kappa}_{\mathcal{S}} = T^{\mathbf{1}}_{\mathcal{S}(\kappa)} \circ \Pi^{\uparrow}_{\kappa}.$$

In particular, we would like to show is that the subsequent diagram commutes for all α:

$$\begin{tikzcd}
x^{\alpha} \arrow[r, mapsto, "T^{\kappa}_{\mathcal{S}}"] \arrow[d, mapsto, "\Pi^{\uparrow}_{\kappa}"'] & g_{\alpha,\mathcal{S}} \arrow[d, phantom, "="] \\
\binom{\kappa}{\alpha}^{-1} e_{\alpha}(x_{\kappa}) \arrow[r, mapsto, "T^{\mathbf{1}}_{\mathcal{S}(\kappa)}"'] & \binom{\kappa}{\alpha}^{-1} \sum_{\widetilde{\alpha} \in e_{\alpha}(x_{\kappa})} g_{\widetilde{\alpha},\mathcal{S}(\kappa)}
\end{tikzcd}$$

where when we write $\widetilde{\alpha} \in e_{\alpha}(x_{\kappa})$, we mean $\widetilde{\alpha} \in supp(e_{\alpha}(x_{\kappa}))$. This holds if we prove the following lemma:

Lemma 2.16. Given any $0 \leq \alpha \leq \kappa$, for any $\widetilde{\alpha} \in e_{\alpha}(x_{\kappa})$:

$$g_{\alpha,\mathcal{S}} = g_{\widetilde{\alpha},\mathcal{S}(\kappa)}.$$

Both equations are exponential generating functions of the images of α and $\widetilde{\alpha}$ along $\mathcal{S}$ and $\mathcal{S}(\kappa)$, respectively. So to prove Theorem 2.16, it suffices to prove that

Lemma 2.17. For any $0 \leq \alpha \leq \kappa$ and any $\widetilde{\alpha} \in e_{\alpha}(x_{\kappa})$, α admits an $\mathcal{S}$-transversal to β if and only if $\widetilde{\alpha}$ admits an $\mathcal{S}(\kappa)$-transversal to β.

Inpsired by the graphical perspective from above, note that we can interpret an $\mathcal{S}$-transversal between α and β, denoted $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$, as a subgraph $\mu \subseteq G_{\mathcal{S}}$ with vertex set $supp(\alpha) \times supp(\beta)$ with $\deg_{\mu}(i) = \alpha_i$ for all $i \in [m]$ and $\deg_{\mu}(j) = \beta_j$ for all $j \in [n]$. Before proving Theorem 2.17, observe that for every $\widetilde{\alpha} \in e_{\alpha}(\kappa)$ we have the following:

Proposition 2.18. For any $\widetilde{\alpha} \in e_\alpha(x_\kappa)$, there exists a pair of maps composing to the identity:

$$\begin{tikzcd} G_{\mathcal{S}} \arrow[r, hook, "\iota_{\widetilde{\alpha}}"] \arrow[rr, bend right, "id"'] & G_{\mathcal{S}(\kappa)} \arrow[r, two heads, "\Pi_\kappa^\downarrow"] & G_{\mathcal{S}} \end{tikzcd}$$

where $\iota_{\widetilde{\alpha}}$ is the obvious embedding and $\Pi_\kappa^\downarrow$ is the contraction map[4].

Proof. For $\widetilde{\alpha} \in e_\alpha(x_\kappa)$, we can present it as $\widetilde{\alpha} = (i_t)$ for some $i_t \in [\kappa_i]$. Then explicitly, the embedding $\iota_{\widetilde{\alpha}} : G_{\mathcal{S}} \to G_{\mathcal{S}(\kappa)}$ is given by sending $i \in [m]$ to $i_t \in [m_\kappa]$, fixing vertices in $[n]$, and sending an edge $(i,j) \in G_{\mathcal{S}}$ to an edge $(i_t, j) \in G_{\mathcal{S}(\kappa)}$. The contraction map $\Pi_\kappa^\downarrow$ contracts all vertices in $[m_\kappa]$ $i_t = i_s$ for fixed $i \in [m]$. □

The final key observation is that subgraphs, in particular those representing transversals, play well with these maps.

Lemma 2.19. Fix $0 \le \alpha \le \kappa$ and $\widetilde{\alpha} \in e_\alpha(\kappa)$. The following hold:

(1) For any $\mu \subseteq G_{\mathcal{S}}$ representing $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$, the subgraph $\iota_{\widetilde{\alpha}}(\mu) \subseteq G_{\mathcal{S}(\kappa)}$ represents $\widetilde{\alpha} \overset{\mathcal{S}(\kappa)}{\longleftrightarrow} \beta$.

(2) For any $\nu \subseteq G_{\mathcal{S}(\kappa)}$ representing $\widetilde{\alpha} \overset{\mathcal{S}(\kappa)}{\longleftrightarrow} \beta$, the subgraph $\Pi_\kappa^\downarrow(\nu) \subseteq G_{\mathcal{S}}$ represents $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$.

Proof. For (1), note that $\iota_{\widetilde{\alpha}}$ does not affect the codomain set $[n]$. In particular, the codomain of $\iota_{\widetilde{\alpha}}(\mu)$ remains β. On the other hand, by definition of the embedding, α maps to $\widetilde{\alpha}$. For (2), the contraction map does not affect the codomain. So β remains after applying $\Pi_\kappa^\downarrow$. On the other hand, by definition of the projection, $\widetilde{\alpha}$ maps to α. □

We the obtain the following cascade of proofs.

Proof of Theorem 2.17. This follows immediately from Theorem 2.19. □

Proof of Theorem 2.16. This is an immediate consequence of Theorem 2.17 □

Proof of Theorem 2.15. Since Theorem 2.16 holds, recalling that there are $\binom{\kappa}{\alpha}$ terms in $e_\alpha(x_\kappa)$, we obtain the desired commutative diagram. □

Proof of Theorem 2.12. Fix $\kappa \in \mathbb{Z}_{\ge 0}^m$. By Theorem 2.15, we can understand $T_{\mathcal{S}}^\kappa$ as a composition of Lorentzian preservers. Since this holds for all κ, $T_{\mathcal{S}}$ is a Lorentzian preserver. □

2.3. **Induction along a set system is a Lorentzian preserver.** In this subsection, we prove that induction along a set system is a Lorentzian preserver by proving that its symbol is Lorentzian. Induction along a set system is similar to transversalization, except we map the normalized monomial $\frac{x^\alpha}{\alpha!}$ instead of the ordinary monomial x^α to the exponential generating function of the image of α.

Let $\mathcal{S} = (S_1, \ldots, S_n)$ be a nonempty set system on $[m]$, i.e., $S_j \neq \varnothing$ for all j. Then, *induction along* $\mathcal{S}$ is the operator

$$\begin{aligned} \mathcal{I}_{\mathcal{S}} : \mathbb{R}[x_1, \ldots, x_m] &\to \mathbb{R}[y_1, \ldots, y_n] \\ \frac{x^\alpha}{\alpha!} &\longmapsto g_\alpha. \end{aligned}$$

[4]Our notation here is again for linguistic determination.

Note that the relationship between $T_{\mathcal{S}}$ and $\mathcal{I}_{\mathcal{S}}$ is a difference in normalization. In the case of matroid stat generating functions, i.e., $\kappa = \mathbf{1}$, they agree. In particular, given a set system $\mathcal{S} = (S_1, \ldots, S_n)$, we define the *induction along* $\mathcal{S}$ to be the map

$$\mathcal{I}_{\mathcal{S}} : \mathbb{R}[x_1, \ldots, x_m] \to \mathbb{R}[y_1, \ldots, y_n]$$

$$\frac{x^\alpha}{\alpha!} \longmapsto g_{\alpha,\mathcal{S}}.$$

Lemma 2.20. Let N be the normalization map. For any set system $\mathcal{S}$,

$$T_{\mathcal{S}} = \mathcal{I}_{\mathcal{S}} \circ N.$$

Since normalization is a Lorentzian preserver, we can also obtain the Lorentzian preserving property of $T_{\mathcal{S}}$ if $\mathcal{I}_{\mathcal{S}}$ is Lorentzian preserving. **DQ says:** We also show this in some other section via geometry, but let us prove it using the symbol theorem.

Immediately observe that the symbol is the generating function of some set:

Observation 2.21. Let $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$ and take $\kappa \in \mathbb{Z}^m_{\geq 0}$. Then, the symbol of induction along $\mathcal{S}$ on domain bounded by κ is:

$$\mathcal{I}^\kappa_{\mathcal{S}}(\mathbf{x}, \mathbf{y}) = \sum_{(\alpha,\beta)\in\mathcal{J}(\mathcal{S})} \frac{\mathbf{x}^{\kappa-\alpha}}{\kappa-\alpha!} \frac{\mathbf{y}^\beta}{\beta!}$$

where $\mathcal{J}(\mathcal{S}) = \{(\alpha, \beta) \in \mathbb{Z}^m \times \mathbb{Z}^n : \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta\}$.

By [Theorem 3.10; BH], it suffices to prove that $supp(\mathcal{I}^\kappa_{\mathcal{S}}(\mathbf{x}, \mathbf{y}))$ is M-convex. Note that the support is a translate by $(\kappa, \mathbf{0})$, so in fact it suffices to prove that

$$\mathfrak{J}(\mathcal{S}) = \{(-\alpha, \beta) : \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta\}$$

is M-convex. We follow the same strategy as in Section 2.1 and prove the following:

Proposition 2.22. The rank function of $\mathfrak{J}(\mathcal{S})$ where $rk_{\mathfrak{J}(\mathcal{S})}(J, T) = \max\{x(J, T) : x \in \mathfrak{J}(\mathcal{S})\}$ for all $(J, T) \subseteq [m] \times [n]$ is submodular.

To do this, we find a nice presentation of the rank function such that submodularity follows suit. The difference from before is now we allow for all subgraphs $\mu \leq G$. We say that a subgraph $\mu \leq G$ has degree (α, β) and notate $\deg(\mu) = (\alpha, \beta)$ if $\deg_\mu(i) = \alpha_i$ and $\deg_\mu(j) = \beta_j$ for $i \in [m]$ and $j \in [n]$. We produce an analogous sequence of cases as in Section 2.1 that lead up to the general statement.

Lemma 2.23. Take some set system $\mathcal{S} = (S_1, \ldots, S_n)$ on $[m]$. Then,

$$rk_{\mathfrak{J}(\mathcal{S})}([m] \times [n]) = 0.$$

Proof. By definition,

$$rk_{\mathfrak{J}(\mathcal{S})}([m] \times [n]) = \max\left\{-\alpha([m]) + \beta([n]) : \alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta\right\}.$$

Notice that for any (α, β) such that $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$, there exists a representative subgraph $\mu \leq G$ with $\deg(\mu) = (\alpha, \beta)$. In particular, notice that

$$-\alpha([m]) + \beta([n]) = -\left(\sum_{i \in [m]} \deg_\mu(i)\right) + \sum_{j \in [n]} \deg_\mu(j) = 0$$

since $\mu \leq G$ is bipartite. □

Next, consider the case when the second coordinate is empty:

Lemma 2.24. Let $J \subseteq [m]$. Then,

$$rk_{\mathfrak{J}(\mathcal{S})}(J, \varnothing) = 0.$$

Proof. Take some $(-\alpha, \beta)$ such that $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$ and some representing subgraph $\mu \leq G$. Then, observe that

$$-\alpha(J) + \beta(\varnothing) = -\left(\sum_{i \in [m]} \deg_\mu(i)\right) \leq 0.$$

Therefore, $rk_{\mathfrak{J}(\mathcal{S})}(J, \varnothing) \leq 0$. On the other hand, the empty subgraph achieves 0. □

Now consider the case where the first coordinate is empty:

Lemma 2.25. Let $T \subseteq [n]$. Then,

$$rk_{\mathfrak{J}(\mathcal{S})}(\varnothing, T) = \sum_{j \in T} |S_j|.$$

Proof. Take some $(-\alpha, \beta)$ such that $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$ and some representing subgraph $\mu \leq G$. Then, observe that

$$-\alpha(\varnothing) + \beta(T) = \sum_{j \in T} \deg_\mu(j).$$

Since $\mu \leq G$, note that $\deg_\mu(j) \leq \deg_G(j) = |S_j|$ for all $j \in [n]$; this gives us the upper bound. Taking the full graph as the subgraph shows that the upper bound is saturated. □

Now we move towards general case, motivated by the cases above. Recall notation from above: $G(J, T)$ is induced subgraph on $J \times T$ and $\mathcal{S}(J, T) = (S_{t_1}|_J, \ldots, S_{t_k}|_J)$ is the induced set system. These agree when we pass between set systems and bipartite graphs.

Lemma 2.26. Let $(J, T) \subseteq [m] \times [n]$. Then,

$$rk_{\mathfrak{J}(\mathcal{S})}(J, T) = rk_{\mathfrak{J}(\mathcal{S}(J^c, T))}(\varnothing, T) = \sum_{j \in T} |S_j|_{J^c}|.$$

Proof. From Theorem 2.25, it suffices to prove that the following bound is tight over $(-\alpha, \beta)$ where $\alpha \overset{\mathcal{S}}{\longleftrightarrow} \beta$, i.e., there exists subgraph $\mu \leq \mathcal{S}$ of degree (α, β):

$$-\sum_{i \in J} \alpha_i + \sum_{j \in T} \beta_j \leq \sum_{j \in T} |S_j|_{J^c}|.$$

Note that $|S_j|_{J^c}| = |S_j| - |S_j|_J|$. As before, we have that $\sum_{j \in T} \beta_j \leq \sum_{j \in T} |S_J|$. So to obtain the inequality, it suffices to prove that $\sum_{i \in J} |S_j|_J| \leq \sum_{i \in J} \alpha_i$. Note that both are enumerating the edges

of the graph $\mathcal{S}(J,[n])$, so in fact, we have equality. To obtain equality, lift the subgraph $\mu \leq \mathcal{S}(J^c,T)$ which maximizes $rk_{\mathfrak{J}(\mathcal{S}(J^c,T))}$ to a subgraph of $\mathcal{S}$. □

Remark 2.27. An important observation for the above proof is that if an edge of a subgraph is supported on J, then it will contribute trivially to the total degree sum. Therefore, we only concern ourselves with edges in the complement of J.

From this presentation, we obtain submodularity:

Proposition 2.28. The rank function $rk_{\mathfrak{J}(\mathcal{S})}$ is submodular.

DQ says: Finish the proof later. But essentially is the following: Submodularity comes from the fact that the (possibly repeating) edges appearing any pair of maximizing subgraphs will cover the edges appearing in the the relevant graph union and intersections.

Remark 2.29. Note that the difference between $\mathfrak{J}(\mathcal{S})$, equivalently $\mathcal{J}(\mathcal{S})$, and $\mathcal{J}_{\mathbf{1}}(\mathcal{S})$ is that now we allow multisets instead of sets. This follows the matroid to polymatroid analogy.

2.4. **Induction along a multigraph is a Lorentzian preserver.** In this subsection, we prove that induction along any bipartite multigraph is a Lorentzian preserver via symbol theory (again). Let $G : V \to W$ be a bipartite multigraph with $|V| = m$ and $|W| = n$. That is, we now allow for multiedges. The set of representatives is then controlled by sub-multigraphs of G. For $\alpha \in \mathbb{Z}^m$ and $\beta \in \mathbb{Z}^n$, we say that α admits a G-transversal to β and write $\alpha \overset{G}{\longleftrightarrow} \beta$ if there exists a sub-multigraph $\mu \leq G$ such that $deg_\mu(V) = \alpha$ and $\deg_\mu(W) = \beta$. Again the symbol is a generating function of some set that could reasonably be M-convex. Namely, the symbol is

$$\mathcal{I}_G^\kappa(\mathbf{x},\mathbf{y}) = \sum_{(\alpha,\beta)\in\mathcal{J}(G)} \frac{\mathbf{x}^{\kappa-\alpha}}{(\kappa-\alpha)!}\frac{\mathbf{y}^\beta}{\beta!}$$

where $\mathcal{J}(G) = \{(\alpha,\beta) : \alpha \overset{G}{\longleftrightarrow} \beta\}$ and we would like to show that

$$\mathfrak{J}(G) = \{(-\alpha,\beta) : \alpha \overset{G}{\longleftrightarrow} \beta\}$$

is M-convex. The ideas leading to the following proposition are the same as above:

Proposition 2.30. Let $(J,T) \subseteq V \times W$. Then,

$$rk_{\mathfrak{J}(G)}(J,T) = rk_{\mathfrak{J}(G|_{J^c\times T})}(\varnothing,T) = \sum_{j\in T} \deg_{G|_{J^c\times T}}(j).$$

Proof. The first equality comes from the fact that any edge supported in J will contribute trivially to the total degree sum. The second equality is proved analogously as above. □

Then submodularity comes more or less for free:

Proposition 2.31. The rank function $rk_{\mathfrak{J}(G)}$ associated to G is submodular.

Proof sketch. **DQ says:** Do the computation at the level of sub-multigraphs. □

This tells us that induction along multigraphs is also Lorentzian preserving.

Theorem 2.32. Let $G : V \to W$ be a bipartite multigraph. Induction along G

$$\mathcal{I}_G : \mathbb{R}[\mathbf{x}_V] \to \mathbb{R}[\mathbf{y}_W], \quad \frac{\mathbf{x}^\alpha}{\alpha!} \longmapsto g_{\alpha,G}$$

where

$$g_{\alpha,G} = \sum_{\alpha \overset{G}{\longleftrightarrow} \beta} \frac{\mathbf{y}^\beta}{\beta!}$$

is a Lorentzian preserver.

2.5. **Induction along a directed (multi)graph is a Lorentzian preserver.** In [Brualdi 71], the author extends induced matroids to arbitrary directed graphs. We do the same for our operator.

Any directed multigraph G on $[n]$ comes equipped with an *outdegree vector* $outdeg_G$ and *indegree vector* $indeg_G$ where $|outdeg_G| = |indeg_G|$. For $\alpha, \beta \in \mathbb{Z}^n$, we will say α admits a G-transversal to β and write $\alpha \xrightarrow{G} \beta$ if there exists a subgraph μ with $outdeg_\mu = \alpha$ and $indeg_\mu = \beta$. We let $\mathcal{J}(G)$ be the set of such pairs:

$$\mathcal{J}(G) = \{(\alpha, \beta) \in \mathbb{Z}^n \times \mathbb{Z}^n : \alpha \xrightarrow{G} \beta\}.$$

This allows us to define *induction along* G as follows:

$$\mathcal{I}_G : \mathbb{R}[x_1, \ldots, x_n] \to \mathbb{R}[x_1, \ldots, x_n]$$
$$\frac{\mathbf{x}^\alpha}{\alpha!} \longmapsto g_{\alpha,G}$$

where $g_{\alpha,G}$ is the exponential generating function of all possible β such that $\alpha \xrightarrow{G} \beta$. So we can ask the question: is this operator Lorentzian preserving?

Our strategy of symbol pushing remains. The symbol calculation is the same as before. So it suffices to prove that the analogous $\mathfrak{J}(G)$ is M-convex. Once again, we turn to finding a good presentation of the rank function. To begin, for any pair of subsets $(S, T) \subseteq [n] \times [n]$:

$$rk_{\mathfrak{J}(G)}(S, T) = \max\{-outdeg_\mu(S) + indeg_\mu(T) : \mu \leq G\}.$$

Recall that for any $S \subseteq [n]$ and $x \in \mathbb{Z}^n$ we use $x(S) \overset{\text{def}}{=} \sum_{i \in S} x_i$. Next, we introduce some graph operations to make sense of the presentation of the rank function.

For any $S \subseteq [n]$ and directed multigraph G on $[n]$ vertices, define $in_S(G)$ to be the subgraph on $[n]$ consisting of the edges pointing *into* S; sometimes we use $in_G(S)$ to emphasize the action on subsets. Analogously, define $out_S(G) = out_G(S)$ to be the subgraph on $[n]$ of edges pointing *out of* S. Let $\overline{out_S}(G) \overset{\text{def}}{=} G \setminus out_S(G)$. Note that $\overline{out_S}(G) = out_{\overline{S}}(G)$ where $\overline{S} = [n] \setminus S$.

Remark 2.33. **DQ says:** Seems like in_S and out_S are functorial on Multigraphs(n).

Observation 2.34. For any $S \subseteq [n]$ and directed multigraph G, by construction,

$$\left|indeg_{in_S(G)}\right| = indeg_G(S) \quad \text{and} \quad \left|outdeg_{out_S(G)}\right| = outdeg_G(S).$$

Now for a series of lemmas:

Lemma 2.35. Let G be a directed multigraph on $[n]$ and $S, T \subseteq [n]$. Then,

$$-outdeg_G(S) + indeg_G(T) \leq indeg_{\overline{out_S}(G)}(T).$$

Proof. Equivalently, we can show that

$$indeg_G(T) - indeg_{\overline{out_S}(G)}(T) \leq outdeg_G(S).$$

Note that for all subgraphs $H \leq G$,

$$indeg_G(T) - indeg_{\overline{H}}(T) = indeg_H(T).$$

In particular, this implies that it suffices to prove that

$$indeg_{out_S(G)}(T) \leq outdeg_G(S).$$

But by the above observation, we have that

$$outdeg_G(S) = \left|outdeg_{out_S(G)}\right| = \left|indeg_{out_S(G)}\right| \geq indeg_{out_S(G)}(T)$$

where the third inequality holds by restriction onto T. □

Lemma 2.36. For any pair of directed multigraphs $\mu \leq G$, for all $S \subseteq [n]$:

$$\overline{out}_S(\mu) \subseteq \overline{out}_S(G).$$

In particular, $indeg_{\overline{out}_S(\mu)} \leq indeg_{\overline{out}_S(G)}$ coordinatewise.

Proof. This is immediate since $\overline{out}_S(G) = out_{\overline{S}}(G)$ and restriction onto $\overline{S} \subseteq [n]$ is order preserving. □

Lemma 2.37. Let $(S, T) \subseteq [n] \times [n]$. Then,

$$rk_{\mathfrak{J}(G)}(S, T) = indeg_{\overline{out}_G(S)}(T).$$

Proof. Fix $S, T \subseteq [n]$. By Theorem 2.35 and Theorem 2.36, we have that $rk_{\mathfrak{J}(G)}(S, T) \leq indeg_{\overline{out_G}(S)}(T)$. The bound is saturated since we maximize over subgraphs and G is a subgraph of itself. □

The submodularity comes via a short calculation.

Proposition 2.38. The rank function $rk_{\mathfrak{J}(G)}$ associated to G is submodular.

Proof. Take $(S, T), (P, Q) \subseteq [n] \times [n]$. We want to show that

$$indeg_{\overline{out_G}(S)}(T) + indeg_{\overline{out_G}(P)}(Q) \geq indeg_{\overline{out_G}(S \cup P)}(T \cup Q) + indeg_{\overline{out_G}(S \cap P)}(T \cap Q).$$

Since $\overline{out_G}(S) = out_G(\overline{S})$, it suffices to prove that

$$indeg_{out_G(S)}(T) + indeg_{out_G(P)}(Q) \geq indeg_{out_G(S \cap P)}(T \cup Q) + indeg_{out_G(S \cup P)}(T \cap Q).$$

Now we show the inequality at the level of graphs since for any graphs G, H on $[n]$

$$indeg_{H \sqcup G} = indeg_H + indeg_G.$$

In particular, we want to show that

$$in_T out_S(G) \sqcup in_Q out_P(G) \supseteq in_{T \cup Q} out_{S \cap P}(G) \sqcup in_{T \cap Q} out_{S \cup P}(G).$$

To shorten notation, lets use $i_T = in_T$ and $o_S = out_S$ and drop G. Note well that for A and B disjoint, $i_{A \sqcup B} = i_A \sqcup i_B$ and analogously for $o_{A \sqcup B} = o_A \sqcup o_B$. Then, note that the left hand side admits the following decomposition:

$$i_T o_S \sqcup i_Q o_P = i_T(o_{S \setminus P} \sqcup o_{S \cap P}) \sqcup (i_{Q \cap T} \sqcup i_{Q \setminus T}) o_P = i_T o_{S \setminus P} \sqcup i_T o_{S \cap P} \sqcup i_{Q \cap T} o_P \sqcup i_{Q \setminus T} o_P.$$

The right hand side of the desired set inequality admits the following decomposition:

$$i_{T\cup Q}o_{S\cap P}\sqcup i_{T\cap Q}o_{S\cup P} = (i_T\sqcup i_{Q\setminus T})o_{S\cap P}\sqcup i_{T\cap Q}(o_P\sqcup o_{S\setminus P}) = i_T o_{S\cap P}\sqcup i_{Q\setminus T}o_{S\cap P}\sqcup i_{T\cap Q}o_P\sqcup i_{T\cap Q}o_{S\setminus P}.$$

The injection should be clear by the coloring. □

Remark 2.39. Notice that, in general, $rk_{\mathfrak{J}(G)}$ is **NOT** nondecreasing. But this is no problem since the actual rank function of interest is $rk_{\mathfrak{J}(G)+(\kappa,0)}(S,T) = rk_{\mathfrak{J}(G)}(S,T) + \kappa(S)$ which is nondecreasing.

Taking $\kappa = outdeg_G(V)$ gives us the desired result:

Theorem 2.40. For any directed multigraph, $\mathcal{I}_G$ is a Lorentzian preserver.

Proof. The symbol $\mathcal{I}_G$ of any bounded degree $\kappa \in \mathbb{Z}^n_{\geq 0}$ is the exponential generating function of $(\kappa, 0) + \mathfrak{J}(G)$. In particular it suffices to prove that $(\kappa, 0) + \mathfrak{J}(G)$ is M-convex. Since M-convexity is translation invariant, it suffices to prove that $\mathfrak{J}(G)$ is M-convex. Equivalently, it suffices to prove that its rank function is submodular. This is the content of Theorem 2.38. □

Remark 2.41. This recovers the result that $\mathcal{I}_G$ is a Lorentzian preserver when $G : [m] \to [n]$ is bipartite. In particular, such bipartite G are directed multigraphs on $[m+n]$ and restriction of the domain is a Lorentzian preserver.

3. Induction is volume preserving, and corollaries

In this subsection, we prove that induction $\mathcal{I}_G$ is volume preserving for G bipartite multigraph. As a corollary, we understand how the operators talk to each other for some pairs of bipartite graphs G and G'. By [Corollary 1.3; GHMSW], it suffices to prove that the symbol is a volume polyonomial. Recall that our symbol is the exponential generating function of a polymatroid $\mathfrak{J}(G)+\kappa$. By [Corollary 1.4; EL23], it suffices to prove that $\mathfrak{J}(G) + \kappa$ is a *linear polymatroid.* We give an explicit characteristic-independent linear representation of the support of $\mathcal{I}_G$, i.e., $\mathfrak{J}(G) + (\kappa, 0)$ when $\kappa = deg_G([m])$.

First, a small lemma to reframe our rank function of $\mathfrak{J}(G) + \kappa$ when $\kappa = deg_G([m])$.

Lemma 3.1. Let rk_G denote the rank function of $\mathfrak{J}(G) + (deg_G([m]), 0) \subseteq \mathbb{Z}^{[n]\times[n]}$. Then the rank of at $(S,T) \subseteq [m] \times [n]$ is the number of edges that are leaving S or entering T:

$$rk_G(S,T) = \#E(out(S) \cup in(T)).$$

Proof. Note that $\deg(S)$ is the number of edges leaving S and $indeg_{out(\overline{S})}(T)$ is the number of edges entering T that are NOT coming from S. □

Now we give the presentation. Consider $G : [m] \to [n]$ bipartite with edge set E, and let N_i be the neighborhoood of $i \in [m]$ and N^j be the neighborhood of $j \in [n]$. The *edge permutation matrix* $E(G) \in \mathbb{R}^{|E|} \times \mathbb{R}^{2|E|}$ is of the form

$$E(G) = \left[\ id \ \middle|\ \sigma(E)\ \right]$$

where $\sigma(E) \in S_E$ is *greedy forgetful permutation*: map from $E = N_1 \sqcup \ldots \sqcup N_m$ and $E = N^1 \sqcup \ldots \sqcup N^n$ and fill greedily along $1 \leq i \leq m$. Equip the edge permutation matrix $E(G)$ with the corresponding

column partition which we call the *neighborhood partition.* Namely, from left to right, give names $N_1, \ldots, N_m, N^1, \ldots, N^n$ with sizes of the neighborhoods. For ease of understanding, label the rows by edges $e \in E$. Then, note that an entry $E(G)_{e,i}$ is nonzero if and only if the edge e is adjacent to N_i or N^i.

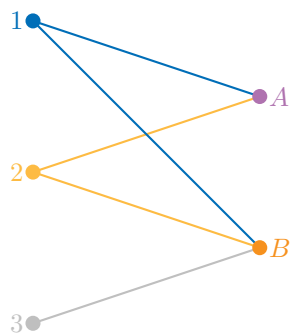

FIGURE 5. Some bipartite G

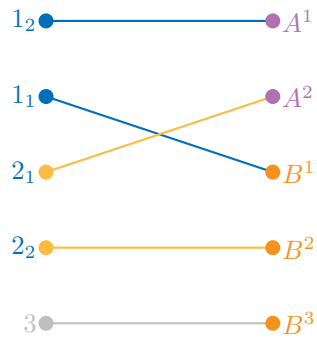

FIGURE 6. greedy forgetful $\sigma(E)$

$$\left(\begin{array}{ccccc|ccccc} 1 & & & & & 1 & & & & \\ & 1 & & & & & & 1 & & \\ & & 1 & & & & 1 & & & \\ & & & 1 & & & & & 1 & \\ & & & & 1 & & & & & 1 \end{array}\right)$$

FIGURE 7. edge permutation matrix $E(G)$

Proposition 3.2. $E(G)$ equipped with its neighborhood partition is a linear representation of the polymatroid $\mathfrak{J}(G) + deg_G([m])$.

Proof. It suffices to prove that the rank of the submatrix $E(G)|_{S\times T}$ is equal to the number of edges facing out of S or into T. Note that the submatrix with columns S is full rank since its a submatrix of the identity. In particular, the number of columns appearing are precisely the number of edges going out of S. On the other hand, the submatrix with columns T is also full rank with rank equal to the number of edges going into T. Now, a column appears in a submatrix S and T if and only if it goes out of S and into T. Hence, the rank is as desired.

DQ says: You can also verify by computing $rk(E(G)_S) + rk(E(G)_T) - rk(E(G)_{S\cap T})$ where $E(G)_{S\cap T}$ is submatrix of columns appearing in both $E(G)_S$ and $E(G)_T$. □

Corollary 3.3. The induction map $\mathcal{I}_G$ is volume preserving.

Proof. Note that for $\kappa > deg_G([m])$ is $\mathcal{I}_G(x^\kappa) = 0$ since there are no subgraphs with $\deg_G([m]) = \kappa$. So $\deg_G([m])$ is a sufficiently large cage to test the symbol. □

This linear presentation plays well with dualization $(-)^\vee$:

Observation 3.4 (Dualization)**.** The dual operator $\mathcal{I}_{G^\vee}$ has the linear presentation

$$E(G^\vee) = [\ id \ \big|\ \sigma(E)^{-1}\ ]$$

equipped with neighborhood partition $N^1, \ldots, N^n, N_1, \ldots, N_m$.

It also plays well with edge deletion, equivalently addition. This implies it plays well other edge operations such as graph unions and intersections; one can easily write down their resulting presentations.

Observation 3.5 (Edge deletion)**.** Given $H = G \setminus e \leq G$ with $E = E(G)$, the edge permutation of H is in $\mathbb{R}^{|E|-1\times 2(|E|-1)}$ and is $E(G)$ where you delete the row labelled E and the remove the resulting zero columns.

Vertex operations like vertex identification can also be described.

Observation 3.6 (Vertex contraction)**.** Let v and w be vertices in the same partite set of bipartite G and let H be the result graph when you identify v and w. Then $E(H) = E(G)$ but now you take the neighborhood partition where you $N_{vw} = N_v \sqcup N_w$.

We also get a different interpretation of $\alpha \to \beta$. Before this, a quick definition. Given a matrix $M \in \mathbb{R}^{m\times n}$ with column partitions $A_1, \ldots, A_r$, we say that a (column) submatrix $N \subseteq M$ is of type $\alpha \in \mathbb{Z}^r_{\geq 0}$ it consists of α_i columns from the A_i-part.

Example 3.7. Take the partition $[123|45|67]$ on the identity matrix. Then the submatrices of type $(2, 0, 1)$ are indexed by $\binom{\{1,2,3\}}{2} \times \{6, 7\}$.

Observation 3.8 (A linear algebra presentation of pairs)**.** Let $G : [m] \to [n]$ be given. Equip the identity with neighborhood partition $N_1, \ldots, N_m$ and the greedy forgetful $\sigma(E)$ with neighborhood partition $N^1, \ldots, N^n$. Then $\alpha \xrightarrow{G} \beta$ if and only if there exists a matrix N that is simultaneously a submatrix of I of type α and a submatrix of $E(G)$ type β.

3.1. **Classifying graphs realizing greedy forgetful.**

Given a permutation σ, can we classify the graphs realizing σ as the greedy forgetful?

Let E be some finte set and $\sigma \in S_E$ be a permutation. Let $\mathsf{GreedyGraphs}(\sigma) \subseteq \mathsf{Multigraphs}$ be the collection of multigraphs realizing σ as the greedy forgetful permutation.

Observation 3.9. The set of greedy multigraphs is in bijection with $Par(E) \times Par(E)$ where $Par(E)$ are labelled partitions, i.e., $[12|3] \neq [1|23]$.

Proof. Take the string diagram of $\sigma \in S_E$ and contract w.r.t to the partition pair. □

DQ says: I think $Par(E)$ should be identified by $(|E|\cdot\triangle_E)/S_E$ where $|E|\cdot\triangle_E \subseteq \left\{\sum_{e\in E} x_e = |E|\right\}$.

- How about the case when we restrict to simple graphs? At the least, it is a saturated subset since simplicity is downward closed. It admits a filtration by #edges allowed per vertex.
- Given $\sigma, \tau \in S_E$ what is $\mathsf{GreedyGraphs}(\sigma) \cap \mathsf{GreedyGraphs}(\tau)$? When is it nonempty?

4. Is induction and/or transveralization stability preserving?

Recall that Lorentzian polynomials were built out of the class of volume polynomials and nonnegative homogeneous real stable polynomials. Since we have proved that induction is a volume preserver, natural question is: "Does induction preserve stability?"

No.

Example 4.1. Consider the set system $\mathcal{S} = (123, 123, 12)$ with domain $\mathbb{R}[a, b, c]$ and codomain $\mathbb{R}[x, y, z]$.

Induction (and transversalization) along $\mathcal{S}$ maps the monomial abc to $\frac{1}{6}((x+y+z)^3 - z^3)$ since the induction along complete set system $K_{3,3}$ takes abc to $\frac{1}{6}(x+y+z)^3$ and the only representative you cannot access via S corresponds to $\frac{z^3}{3!}$. Note that we have a factorization

$$(x+y+z)^3 - z^3 = (x+y)((x+y+z)^2 + (x+y+z)z + z^2) = (x+y)z^2(v^2+v+1)$$

where $v = (x+y+z)/z$. Furthermore $v = e^{\frac{2\pi i}{3}}$ is a root of v^2+v+1 and is in the upper half plane.

So we can construct a solution like $z = e^{\frac{\pi i}{12}}$ and $x = y = \frac{1}{2}(e^{\frac{3\pi i}{4}} - e^{\frac{\pi i}{12}})$; importantly, $\mathfrak{Im}(x, y, z) > 0$, so the image of abc is not stable under induction along $\mathcal{S}$. In particular, the transversalized basis generating polynomials for graphic matroids are not stable in general.

Remark 4.2. There are infinitely many such G where $\mathcal{I}_G$ fails to be stability preserving.

Construction #1. Recall that the Catalan matroid C_n has bases consisting of up-steps for each Dyck path of length $2n$ [Catalan Matroids, Ardila]. In particular, it has a presentation by the set system $\mathcal{C}_n^\vee = ([1], [3], [5], ..., [2n-1])$. Now consider

$$\mathcal{I}_{\mathcal{C}_n} : \mathbb{R}[x_1, \ldots, x_n] \to \mathbb{R}[y_1, \ldots, y_n]$$

and observe that $(MAP \circ \mathcal{I}_{\mathcal{C}_n})(x_1, \ldots, x_n) = BGP(C_n)$. By [Theorem 2.6, Alexandersson-Jal], $BGP(C_n)$ is not stable for $n \geq 10$. In particular, since MAP preserves stability, $\mathcal{I}_{\mathcal{C}_n}$ is not a stability preserver. □

Construction #2. From [Proposition 5.9, Choe-Wagner], there exists a set system $\mathcal{S}^\vee$ consisting of four sets on $[12]$ whose transversal matroid fails to be *balanced.* In the same paper, they prove that HPP[5] implies Rayleigh implies balanced, so in particular, $\mathcal{I}_{\mathcal{S}^\vee}(x^{[4]})$ is not stable. Recall that stability is a minor closed property so for any set system $\mathcal{T}$ on $[m] \supseteq [12]$, $\mathcal{I}_{\mathcal{S}^\vee \oplus \mathcal{T}}$ is not stablility preserving. □

[5]A matroid is HPP if its BGP is stable.